

Cem Alpaslan's Master's Project - Mathematical Model

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1 Mathematical Modelling

1.1 Assumptions

- It is assumed that there will be no split deliveries in this mathematical formulation.
- It is assumed that all vehicles must turn back to depot at the end of the day.(Not an Open VRP).
- It is assumed that at the end of each trip and each day, vehicle turns back to the depot and may partially load and unload its cargo. Cross-docking is allowed in this problem.

1.2 Sets

- $T = \{1, 2\}$: Set of days
- $K = \{1, 2, 3\}$: Set of vehicles, 1 is small capacity vehicle, 2 is medium capacity vehicle and 3 is large capacity vehicle
- $R = \{1, 2, \dots, R_{\max}\}$: Set of trips where $R_{\max} = |P| + |D|$
- $N = \{0\} \cup P \cup D$: Set of nodes where 0 denotes the depot, P pickup nodes, D delivery nodes
- $A = \{(i, j) \in N \times N : i \neq j\}$: Set of arcs
- $S =$ Set of shipments where $s \in S$, $p(s) \in P$ is the pickup node and $d(s) \in D$ is the delivery node

1.3 Parameters

- D_{ij} : Distance from node i to node j
- T_{ij} : Travel time from node i to node j
- S_i : Service duration time at node i
- E_i : Earliest service start time at node i
- L_i : Latest service start time at node i
- E_{0t} : Earliest opening time of the depot on day t
- L_{0t} : Latest closing time of the depot on day t
- Q_k : Cargo capacity of vehicle k
- C^{dep} : Storage capacity of the depot
- F_k : Fixed daily rental cost of vehicle k
- A_k : Distance per km cost (fuel and driver wage) for vehicle k
- H_{kt} : Maximum daily travel distance limit of vehicle k on day t
- τ : Reload/handling time at depot between two trips
- q_s : Amount of shipment s
- C^{cd} : Cost incurred when a shipment changes vehicles at the depot (cross-docking cost)
- I_0 : Initial inventory level at the depot before the shipment horizon
- $\alpha_s \in \{0, 1\}$: Pickup status parameter
 - $\alpha_s = 1$: Shipment already picked up before the horizon and ready in the depot ($p(s) = \{0\}$)
 - $\alpha_s = 0$: Pickup must be done within the horizon ($p(s) \neq \{0\}$)
- Δ_i : Net load change at node i

$$\Delta_i = \sum_{s: i=p(s)} q_s - \sum_{s: i=d(s)} q_s$$

pickup node: $\Delta_i > 0$ (load increase)

delivery node: $\Delta_i < 0$ (load decrease)
- M : Large positive constant (Big-M)

1.4 Decision Variables

- $x_{ij}^{ktr} \in \{0, 1\}$: 1 if vehicle k travels from node i to node j on trip r on day t ; 0 otherwise
- $w^{ktr} \in \{0, 1\}$: 1 if trip r of vehicle k is active on day t ; 0 otherwise
- $v_{kt} \in \{0, 1\}$: 1 if vehicle k is rented on day t ; 0 otherwise
- $b_i^{ktr} \geq 0$: Service start time at node i for vehicle k on trip r on day t
- $u_i^{ktr} \geq 0$: Load of vehicle k after leaving node i on trip r on day t
- $l_{\text{out}}^{ktr} \geq 0$: Load taken from the depot at the *start* of trip r by vehicle k on day t
- $l_{\text{in}}^{ktr} \geq 0$: Load dropped off at the depot at the *end* of trip r by vehicle k on day t
- $I_t \geq 0$: Inventory level at the depot at the end of day t , $\forall t \in T$
- $\delta_s \in \{0, 1\}$: 1 if shipment s changes vehicles at the depot (cross-docking); 0 otherwise

1.5 Objective Function

$$\min Z = \sum_{t \in T} \sum_{k \in K} F_k v_{kt} + \sum_{t \in T} \sum_{k \in K} \sum_{r \in R} \sum_{(i,j) \in A} A_k D_{ij} x_{ij}^{ktr} + C^{\text{cd}} \sum_{s \in S} \delta_s \quad (1)$$

The objective function minimizes the total cost including: daily vehicle rental costs, distance-based transportation costs, and cross-docking(load exchange) transfer costs at the depot.

1.6 Constraints

$$w^{ktr} \leq v_{kt} \quad \forall k \in K, t \in T, r \in R \quad (2)$$

Vehicle should be hired if a trip is active. This constraint avoids creating a trip with an unrented vehicle.

$$w^{ktr} \leq w^{ktr'} \quad \forall k \in K, t \in T, r, r' \in R : r' < r \quad (3)$$

This constraint eliminates symmetric solutions by enforcing sequential trip activation. If trip r is active, then all trips r' with $r' \leq r$ must also be active, it prevents the solver from generating equivalent solutions with different trip index combinations.

$$\sum_{j \in N \setminus \{0\}} x_{0j}^{ktr} = w^{ktr} \quad \forall k \in K, t \in T, r \in R \quad (4)$$

If a trip is active there should be one exit from the depot. This constraint ensures that the trip will start from the depot.

$$\sum_{i \in N \setminus \{0\}} x_{i0}^{ktr} = w^{ktr} \quad \forall k \in K, t \in T, r \in R \quad (5)$$

If a trip is active there should be one return to the depot. This constraint ensures that the trip will end at the depot.

$$\sum_{j \in N} x_{ij}^{ktr} - \sum_{j \in N} x_{ji}^{ktr} = 0 \quad \forall i \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (6)$$

This constraint enforces route continuity by balancing the incoming and outgoing flows at each visited node. If a vehicle enters a node, it must also leave that node during the same trip.

$$\sum_{k \in K} \sum_{t \in T} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} = 1 \quad \forall s \in S : \alpha_s = 0 \quad (7)$$

This constraint ensures that if the pickup of shipment s has not been completed before the planning horizon ($\alpha_s = 0$), its pickup node must be visited exactly once within the horizon.

$$\sum_{k \in K} \sum_{t \in T} \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} = 1 \quad \forall s \in S \quad (8)$$

With this constraint delivery of shipment s must be fulfilled by visiting the delivery node ($d_s = 1$) exactly once.

$$0 \leq l_{out}^{ktr} \leq Q_k w^{ktr} \quad \forall k \in K, t \in T, r \in R \quad (9)$$

This constraint ensures that the load taken from the depot at the start of the trip cannot exceed the vehicle capacity.

$$0 \leq l_{in}^{ktr} \leq Q_k w^{ktr} \quad \forall k \in K, t \in T, r \in R \quad (10)$$

This constraint ensures that the load dropped off to the depot at the end of the trip cannot exceed the vehicle capacity.

$$u_0^{kt1} = l_{out}^{kt1} \quad \forall k \in K, t \in T \quad (11)$$

At the start of the first trip of each day, the vehicle load equals the amount loaded from the depot. The vehicle begins the day empty and takes every pallet from the depot.

$$u_0^{ktr} = l_{out}^{ktr} \quad \forall k \in K, t \in T, r \in R : r > 1 \quad (12)$$

This constraint ensures that the depot departure load equals the load taken from the depot at the start of each trip $r > 1$.

$$\sum_{k \in K} \sum_{r \in R} u_0^{ktr} = \sum_{k \in K} \sum_{r \in R} u_0^{k(t-1)r} + \sum_{k \in K} \sum_{r \in R} \sum_{i \in N} \sum_{j \in N \setminus \{i\}} \Delta_j \cdot x_{ij}^{k(t-1)r} - \sum_{k \in K} \sum_{r \in R} l_{in}^{k(t-1)r} + \sum_{k \in K} \sum_{r \in R} l_{out}^{ktr} \quad \forall t \in T \quad (13)$$

This constraint ensures the daily aggregate depot departure load balance across all vehicles and trips. The total load departing the depot on day t equals the previous day's total departure load, adjusted for net pickup and delivery changes, minus loads returned to the depot on day $t-1$, plus loads collected from the depot on day t .

$$u_j^{ktr} \geq u_0^{ktr} + \Delta_j - M(1 - x_{0j}^{ktr}) \quad \forall j \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (14)$$

This constraint ensures the lower bound for load update after the first pickup or delivery following depot departure. This constraint ensures the load update for depot to node scenario.

$$u_j^{ktr} \leq u_0^{ktr} + \Delta_j + M(1 - x_{0j}^{ktr}) \quad \forall j \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (15)$$

This constraint ensures the upper bound for load update after the first pickup or delivery following depot departure. This constraint ensures the load update for depot to node scenario.

$$u_j^{ktr} \geq u_i^{ktr} + \Delta_j - M(1 - x_{ij}^{ktr}) \quad \forall (i, j) \in A : j \neq 0, k \in K, t \in T, r \in R \quad (16)$$

This constraint satisfies the lower bound of load update along the route considering pickups and deliveries. This constraint ensures the load update for node to node scenario.

$$u_j^{ktr} \leq u_i^{ktr} + \Delta_j + M(1 - x_{ij}^{ktr}) \quad \forall (i, j) \in A : j \neq 0, k \in K, t \in T, r \in R \quad (17)$$

This constraint ensures the upper bound of load update along the route considering pickups and deliveries. This constraint ensures the load update for node to node scenario.

$$u_i^{ktr} - M(1 - x_{i0}^{ktr}) \leq l_{in}^{ktr} \leq u_i^{ktr} + M(1 - x_{i0}^{ktr}) \quad \forall i \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (18)$$

This constraint ensures that the load deposited at the depot equals the vehicle load at the last visited node before returning to the depot.

$$0 \leq u_i^{ktr} \leq Q_k w^{ktr} \quad \forall i \in N, k \in K, t \in T, r \in R \quad (19)$$

This constraint ensures the general capacity. Vehicle load cannot exceed the vehicle capacity at any node, including the depot.

$$u_j^{ktr} \leq Q_k \cdot \sum_{i \in N} x_{ij}^{ktr} \quad \forall j \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (20)$$

This constraint ensures that the load of vehicle k at node j on trip r on day t is zero if the vehicle does not visit node j . It forces the load to be zero.

$$b_j^{ktr} \geq b_i^{ktr} + S_i + T_{ij} - M(1 - x_{ij}^{ktr}) \quad \forall (i, j) \in A : j \neq 0, k \in K, t \in T, r \in R \quad (21)$$

With this constraint, service start times must follow travel and service durations.

$$E_i - M \left(1 - \sum_{j \in N} x_{ij}^{ktr} \right) \leq b_i^{ktr} \leq L_i + M \left(1 - \sum_{j \in N} x_{ij}^{ktr} \right) \quad \forall i \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (22)$$

This constraint ensures that service must start within the time window of node i if the node is visited.

$$E_{0t} \cdot w^{ktr} \leq b_0^{ktr} \leq L_{0t} \cdot w^{ktr} \quad \forall k \in K, t \in T, r \in R \quad (23)$$

The depot departure time of every active trip must fall within the depot's operating hours on day t . It ensures the depot time window restrictions.

$$b_0^{ktr'} \geq b_i^{ktr} + S_i + T_{i0} + \tau - M \left(2 - x_{i0}^{ktr} - w^{ktr'} \right) \quad \forall i \in N \setminus \{0\}, k \in K, t \in T, r, r' \in R : r' > r \quad (24)$$

This constraint ensures correct trip sequencing. The depot departure time of trip r' must be no earlier than the actual return time of trip r , including the service time and travel time from the last visited node i to the depot, plus the handling time τ .

$$b_0^{ktr} + \sum_{i \in N \setminus \{0\}} S_i \sum_{j \in N} x_{ij}^{ktr} + \sum_{i \in N} \sum_{j \in N} T_{ij} x_{ij}^{ktr} \leq L_{0t} \quad \forall k \in K, t \in T, r \in R \quad (25)$$

This constraint ensures that the total time of trip r , including depot departure time, service durations, and travel times across all visited nodes, does not exceed the depot closing time L_{0t} on day t .

$$\sum_{t \in T} t \cdot \sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \leq \sum_{t \in T} t \cdot \sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \quad \forall s \in S : \alpha_s = 0 \quad (26)$$

This constraint ensures that the pickup of shipment s is performed on a day no later than its delivery day. It is a predecessor constraint.

$$b_{p(s)}^{ktr} \leq b_{d(s)}^{ktr} + M \left(2 - \sum_{j \in N} x_{p(s),j}^{ktr} - \sum_{j \in N} x_{d(s),j}^{ktr} \right) \quad \forall s \in S : \alpha_s = 0, k \in K, t \in T, r \in R \quad (27)$$

This constraint ensures that when the pickup and delivery of shipment s are served by the same vehicle on the same trip, the pickup node is visited before the delivery node.

$$b_0^{k_{atrd}} \geq b_i^{k_{ptrp}} + S_i + T_{i,0} - M \left(1 - x_{i,0}^{k_{ptrp}} \right) - M \left(2 - \sum_{j \neq p(s)} x_{p(s),j}^{k_{ptrp}} - \sum_{j \neq d(s)} x_{d(s),j}^{k_{atrd}} \right) \quad (28)$$

$$\forall s \in S : \alpha_s = 0, \quad i \in N \setminus \{0\}, \quad t \in T, \quad k_p, k_d \in K, \quad r_p, r_d \in R$$

This constraint enforces same-day cross-docking temporal ordering. When shipment s is picked up and delivered on the same day by different trips, the delivery trip must not depart the depot before the pickup trip has returned. The Big-M terms deactivate the constraint when the relevant arcs or nodes are not active on their respective trips.

$$\sum_{r \in R} \sum_{(i,j) \in A} D_{ij} x_{ij}^{ktr} \leq H_{kt} \quad \forall k \in K, t \in T \quad (29)$$

This constraint ensures that total daily distance traveled by a vehicle cannot exceed its daily limit.

$$I_0 = \sum_{s \in S} \alpha_s q_s \quad (30)$$

This constraint ensures the initial inventory that has already been picked up and stored at the depot before the planning horizon.

$$I_t = I_{t-1} + \sum_{s \in S : \alpha_s = 0} q_s \sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} - \sum_{s \in S} q_s \sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \quad \forall t \in T \quad (31)$$

This constraint ensures the daily inventory balance at the depot considering pickups and deliveries performed during day t .

$$\sum_{k \in K} \sum_{r \in R} l_{out}^{ktr} \leq I_{t-1} + \sum_{s \in S : \alpha_s = 0} q_s \sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \quad \forall t \in T \quad (32)$$

This constraint ensures that the total load dispatched from the depot on day t cannot exceed the available inventory (previous day closing stock plus pickups made on day t).

$$I_t \leq C^{dep} \quad \forall t \in T \quad (33)$$

This constraint ensures that the depot inventory level at the end of each day t does not exceed the depot's storage capacity.

$$\delta_s \geq \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} - \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \quad \forall s \in S : \alpha_s = 0, k \in K, t \in T \quad (34)$$

$$\delta_s \geq \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} - \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \quad \forall s \in S : \alpha_s = 0, k \in K, t \in T \quad (35)$$

These two constraints force $\delta_s = 1$ whenever the pickup and delivery of shipment s are performed under different conditions for vehicle k on day t . They capture the following cross-docking scenario: same day with different vehicles.

$$\delta_s \geq \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} - \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{kt'r'} \quad \forall s \in S : \alpha_s = 0, k \in K, t \in T, t' \in T : t' > t \quad (36)$$

$$\delta_s \geq \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{kt'r'} - \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \quad \forall s \in S : \alpha_s = 0, k \in K, t \in T, t' \in T : t' > t \quad (37)$$

These two constraints force $\delta_s = 1$ whenever the pickup and delivery of shipment s are performed under different conditions for vehicle k on day t . They capture the following cross-docking scenarios: different day with same vehicle, and different day with different vehicles.

$$\delta_s \leq 2 - \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} - \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \quad \forall s \in S : \alpha_s = 0, k \in K, t \in T \quad (38)$$

This constraint ensures that ($\delta_s = 0$) whenever the pickup and delivery of shipment s are performed by the same vehicle k on any of its trips r on day t .

$$l_{\text{in}}^{ktr} \geq q_s \sum_{j \in N} x_{p(s),j}^{ktr} - M(1 - \delta_s) \quad \forall s \in S, k \in K, t \in T, r \in R \quad (39)$$

This constraint ensures that if a pickup of shipment s is made and cross-docking will be performed ($\delta_s = 1$), then the load must be unloaded to the depot at the end of that trip so that another vehicle can collect and deliver it.

$$l_{\text{out}}^{ktr} \geq q_s \sum_{j \in N} x_{d(s),j}^{ktr} - M(1 - \delta_s) \quad \forall s \in S, k \in K, t \in T, r \in R \quad (40)$$

This constraint ensures that if the delivery of shipment s is performed and cross-docking will be performed ($\delta_s = 1$), then the delivering vehicle must have collected at least the cross-docked load from the depot at the start of that trip.

$$l_{\text{out}}^{ktr} \geq \sum_{s \in S : \alpha_s = 1} q_s \sum_{j \in N} x_{d(s),j}^{ktr} \quad \forall k \in K, t \in T, r \in R \quad (41)$$

This constraint ensures that when delivering pre-positioned shipments ($\alpha_s = 1$), the vehicle must have loaded at least the corresponding quantity from the depot at the start of that trip.

$$\delta_s = 0 \quad \forall s \in S : \alpha_s = 1 \quad (42)$$

This constraint ensures that cross-docking is not applicable for pre-positioned shipments since they are already at the depot and do not require a pickup vehicle.

Domain Constraints

$$x_{ij}^{ktr} \in \{0, 1\} \quad \forall (i, j) \in A, k \in K, t \in T, r \in R \quad (43)$$

$$w^{ktr} \in \{0, 1\} \quad \forall k \in K, t \in T, r \in R \quad (44)$$

$$v_{kt} \in \{0, 1\} \quad \forall k \in K, t \in T \quad (45)$$

$$\delta_s \in \{0, 1\} \quad \forall s \in S \quad (46)$$

$$b_i^{ktr} \geq 0 \quad \forall i \in N, k \in K, t \in T, r \in R \quad (47)$$

$$u_i^{ktr} \geq 0 \quad \forall i \in N, k \in K, t \in T, r \in R \quad (48)$$

$$l_{\text{out}}^{ktr} \geq 0 \quad \forall k \in K, t \in T, r \in R \quad (49)$$

$$l_{\text{in}}^{ktr} \geq 0 \quad \forall k \in K, t \in T, r \in R \quad (50)$$

$$I_t \geq 0 \quad \forall t \in T \quad (51)$$